

The Final Presentation

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Disclaimer:
Please follow the advice, not this example of
presentation

Key Intuition

The audience

I know... this might seem depressing after all your efforts...

The audience

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But your committee is made of (multiple options apply):

- Domain Experts
- Experienced Professors working in completely different domains
- Bored / Tired / Distracted Professors for little hours slept

You want the committee to easily understand you have been doing valuable things.... without having to be:

- Fully focused on what you are telling them
- Completely aware of your background

Stick to Standards!

- Your presentation has to be **self-contained**
- You cannot blame us if we miss some minor detail in a single slide (after all, it's us grading you, not the other way round...)
- Having a «**standard**» **structure** can help your audience attend your presentation.
- Avoid further distracting elements... we get easily distracted from «outliers».
- Use both texts and nice illustrations/schemes.

There are two groups in the audience: the
grannies and the *pros*



Grannies

These do not have specific knowledge in computer vision/machine learning.

You have to be so kind to **let everybody in Grannies group to understand:**

- what is the problem you have been addressing and why that was challenging
- what is your (brilliant) idea to solve the problem
- whether you have been successful and how
- the key advantages of your solution



Pros

These "pros" can **perfectly understand whether you are a serious guy or you are just cheating**. You have to clearly explain to these:

- your actual solution
- the major advantages of your solution
- the figures of merit you adopted for the experiments

The pro group might contain professionals with a very specific technical background, and you have to **meet their expectations** as well.



Presentation Structure

Important

- You should not report everything you have been doing
- You should not follow a chronological order
- You should **polish all your work in a concise and effective narrative**
- You should **enthusiastically convey your key findings!**

Duration: 15-minute talk for a thesis, tesina is typically shorter (say 10 minutes),

Introduction

Clearly describe:

- the problem you have been addressing (the context and the specific problem),
- the reason why you have been studying this (namely, why it is an important problem),
- the **rationale behind your solution**
- what makes your solution original and novel.

When you illustrate your idea, ensure that everyone in the audience can understand it.

Problem Formulation

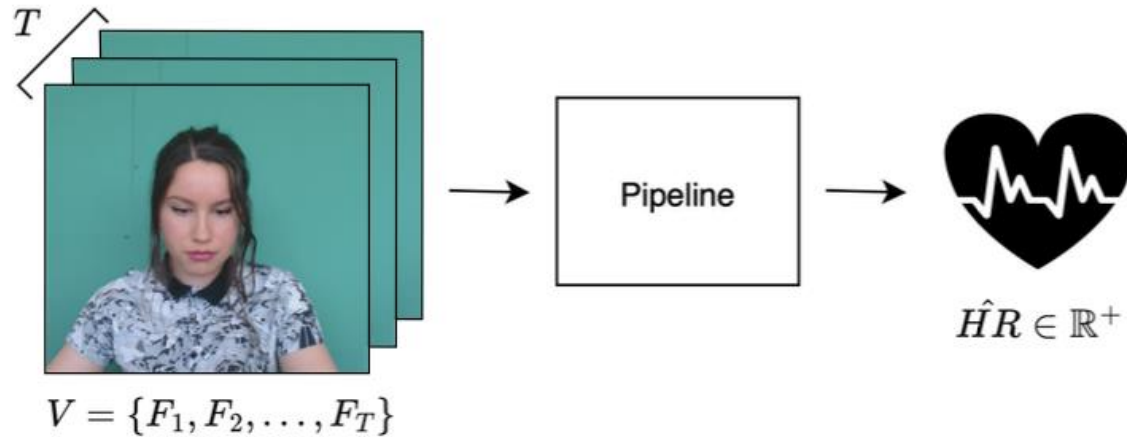
A more compact version than that in the executive summary

Please include:

- **Formulas**, but make sure all these are needed and each of these is being presented.
- **Illustrations**: that's definitively a plus, can really speed up illustrating the problem formulation

Problem Formulation

Problem Formulation



Goal: estimate the Heart Rate of the recorded subject



Input: $V = \{F_1, F_2, \dots, F_T\}$ where F_i is the i -th frame of the video V

Output: $\hat{HR} \in \mathbb{R}^+$ the final Heart Rate estimate

Problem formulation

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Template Matching

Input

- Reference Image

$$X \in [0,255]^{W \times H \times 3}$$

- Templates

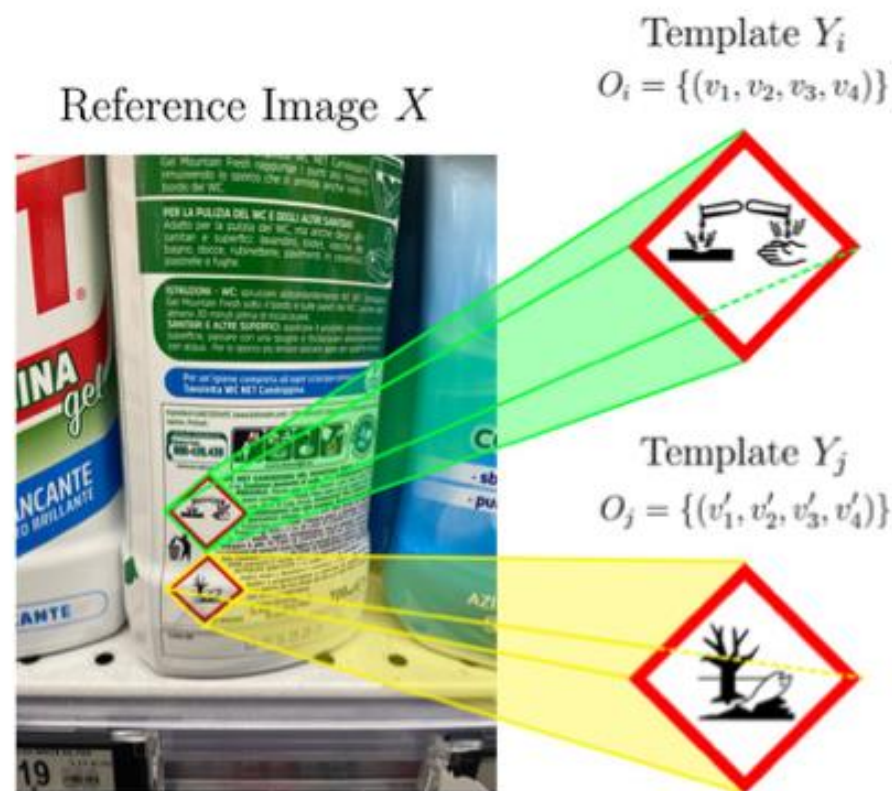
$$Y_m \in [0,255]^{w_m \times h_m \times 3}$$

with a quadrilateral outline and
a small number of plain colors

Output

- Found instances per template

$$O_m = \{(v_1, v_2, v_3, v_4)\}_{i=1}^{N_m}$$



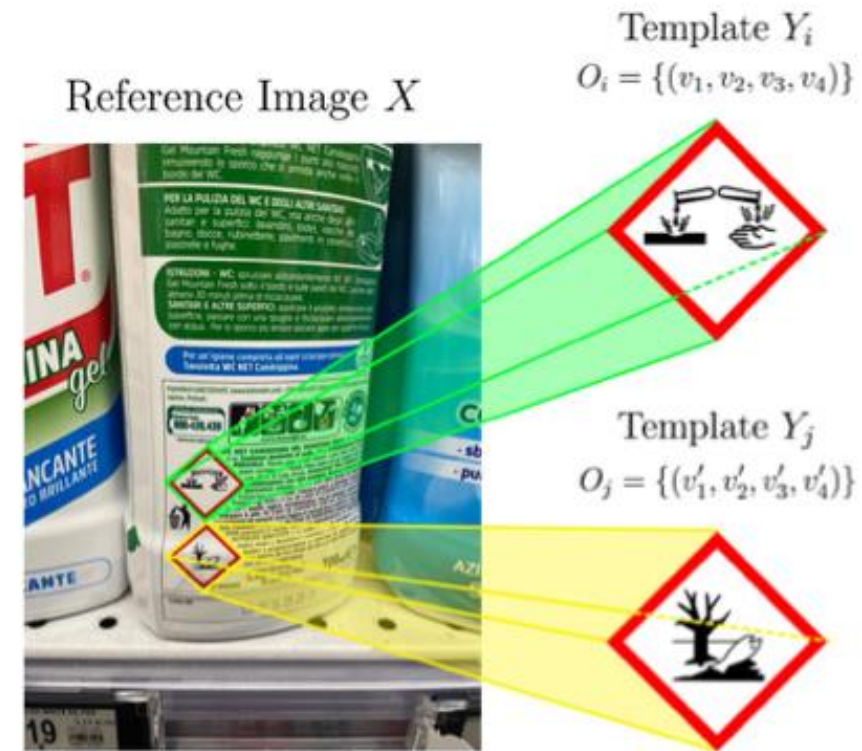
Problem formulation

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Template Matching: Goals

- Minimize false positives and false negatives
- Work with **no model training** or annotated dataset.
No retraining when changing templates.
- Run on **any device** without GPU
- Provide **real-time** performances

Mention also the “rules of the game”

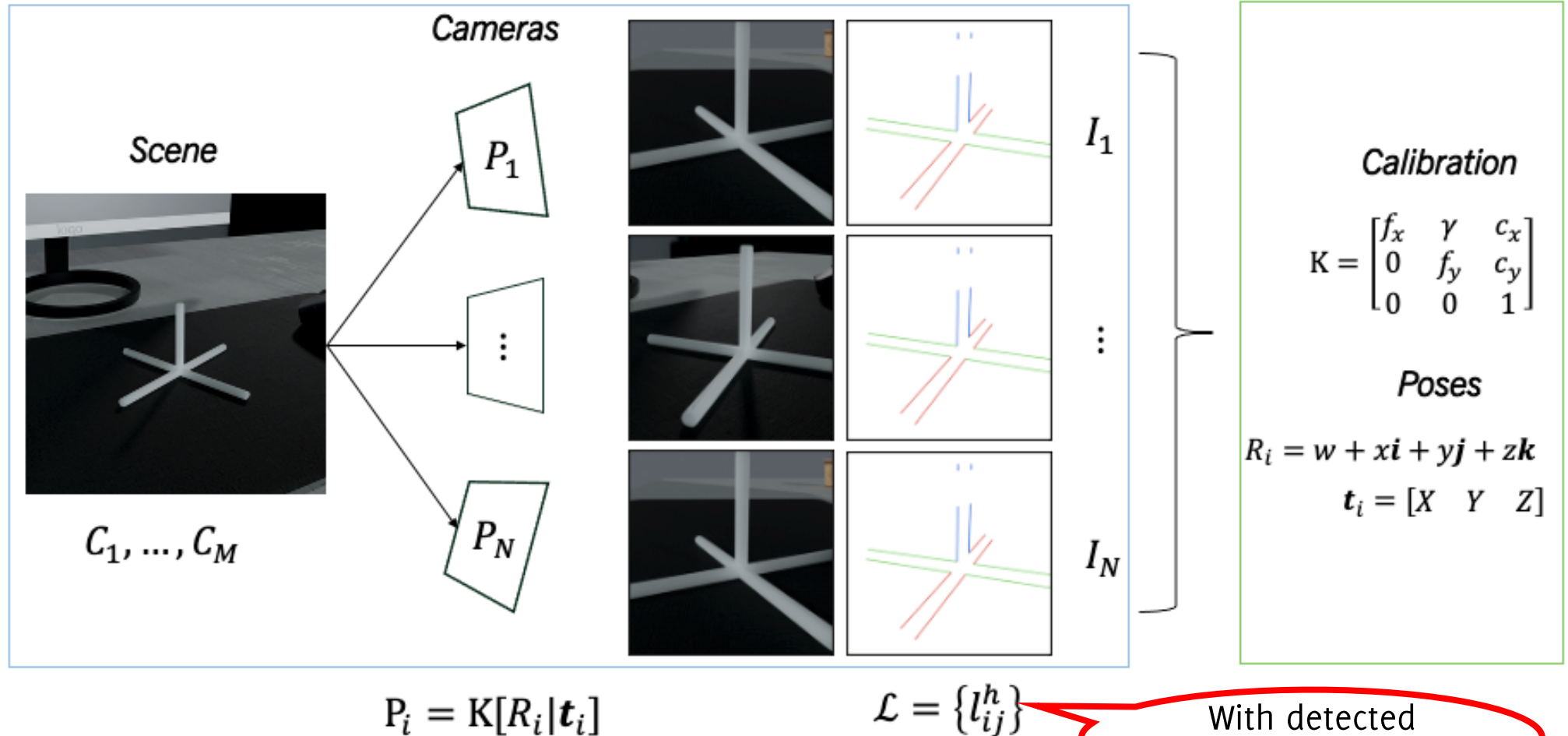


Problem Formulation

Anyone jumping in at any moment, should be able to understand the content of the slide, **write all the necessary details**

of a scene with 3D known cylinders... make explicit all the assumptions

PROBLEM FORMULATION



With detected silhouette lines

Related Work

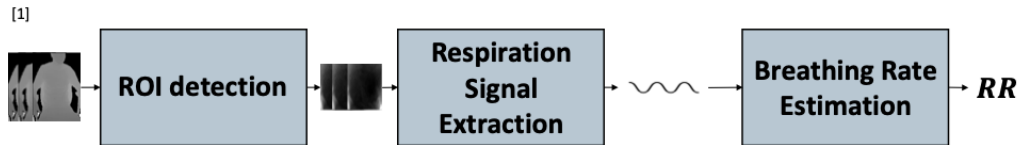
Two purposes:

- **you show you master the scientific literature/context** of your solution
- emphasizing the limitations of the current solutions that you can overcome with your method.

DEPTH CAMERA-BASED METHODS



State of the art pipeline



[1] Wichum, F., Wiede, C., and Seidl, K.: Depth-Based Measurement of Respiratory Volumes: A Review. Sensors (Basel, Switzerland). 22(24):9680. December 2022

DEPTH CAMERA-BASED METHODS



State of the art methods struggle with **noise**, **occlusions**, and **artifacts** that must be handled **explicitly**



Neural Networks learn to manage complexities **implicitly**



No current depth-based neural network reconstructs **respiratory signal** or predict its **rate**

Related Work

.. but maybe you can illustrate this **without a dedicated slide.**

- in the core contribution slide, you can mention that you took [XX] from [YY]
- In the experiments, you mention that you compare against methods that are [ZZ],[WW]

Core Contribution

Then, present your **core contribution** in a technical manner.

Again, use formulas and schemes as long as these are discussed

There is no need (better to avoid!) naming the section “Core Contribution”: use the name of your algorithm/solution “e.g., BM₃D” or what it does “removing signal-dependent noise”

Overview of the method

Try to convey the key ideas at high level.

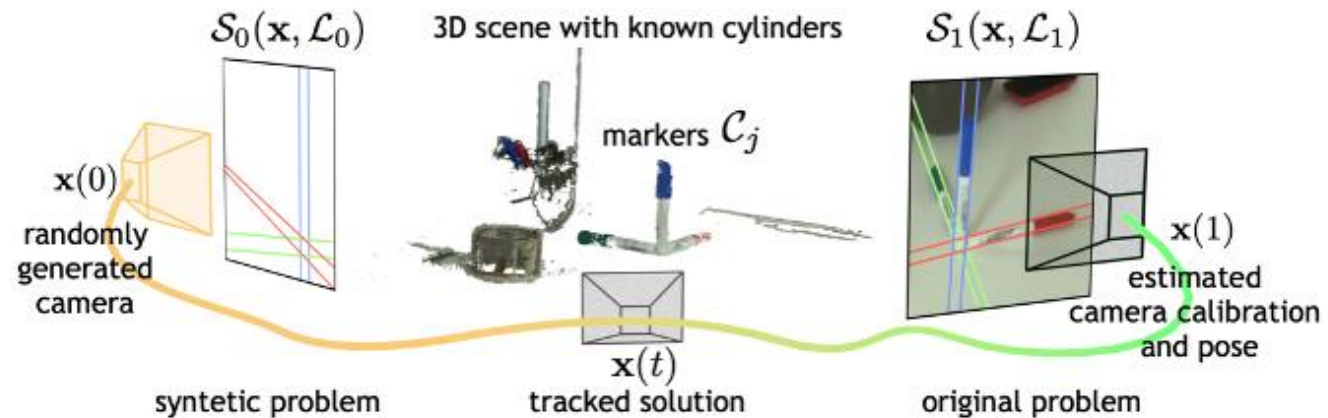
CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION

Equation arising from quadric geometry are quartic equations, each with multiple solutions. The **problem** can be **projected** into a higher space in which is linear but is both **computationally intensive**, and **geometrically inconsistent**.

My solution aims to **solve both issues by deforming** an easy, **known** version of the **problem** into the hard, unknown, **target problem**.

$$\begin{cases} l^h{}^T P Q^* P^T l^h = 0 \\ l^h{}^T P V = 0 \end{cases}$$

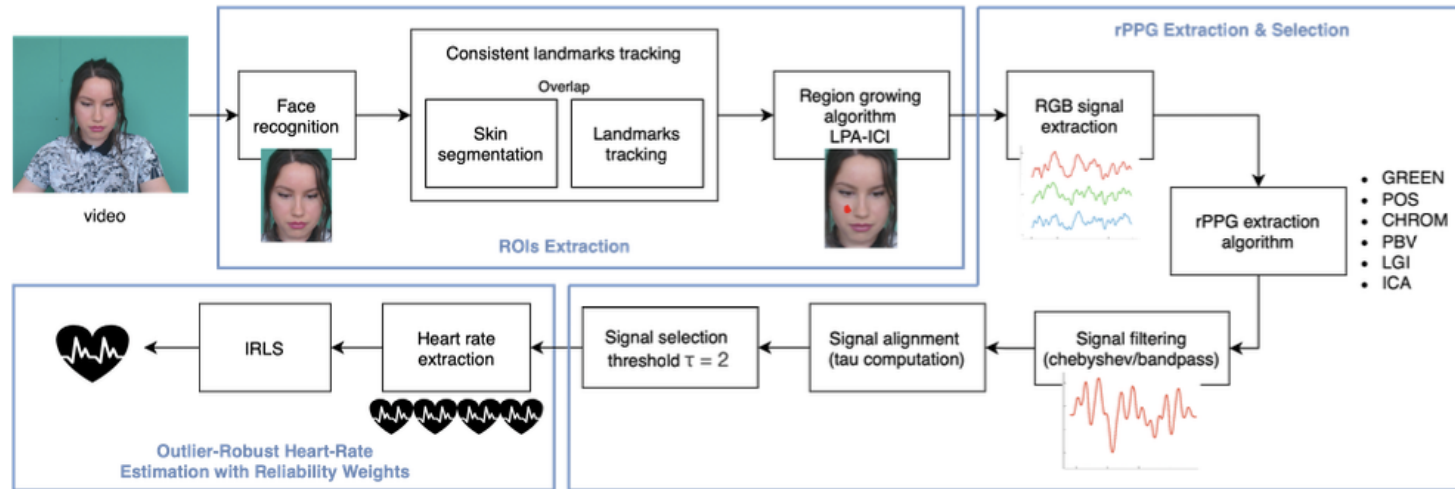
Use formulas



Use illustrations

Overview of the method

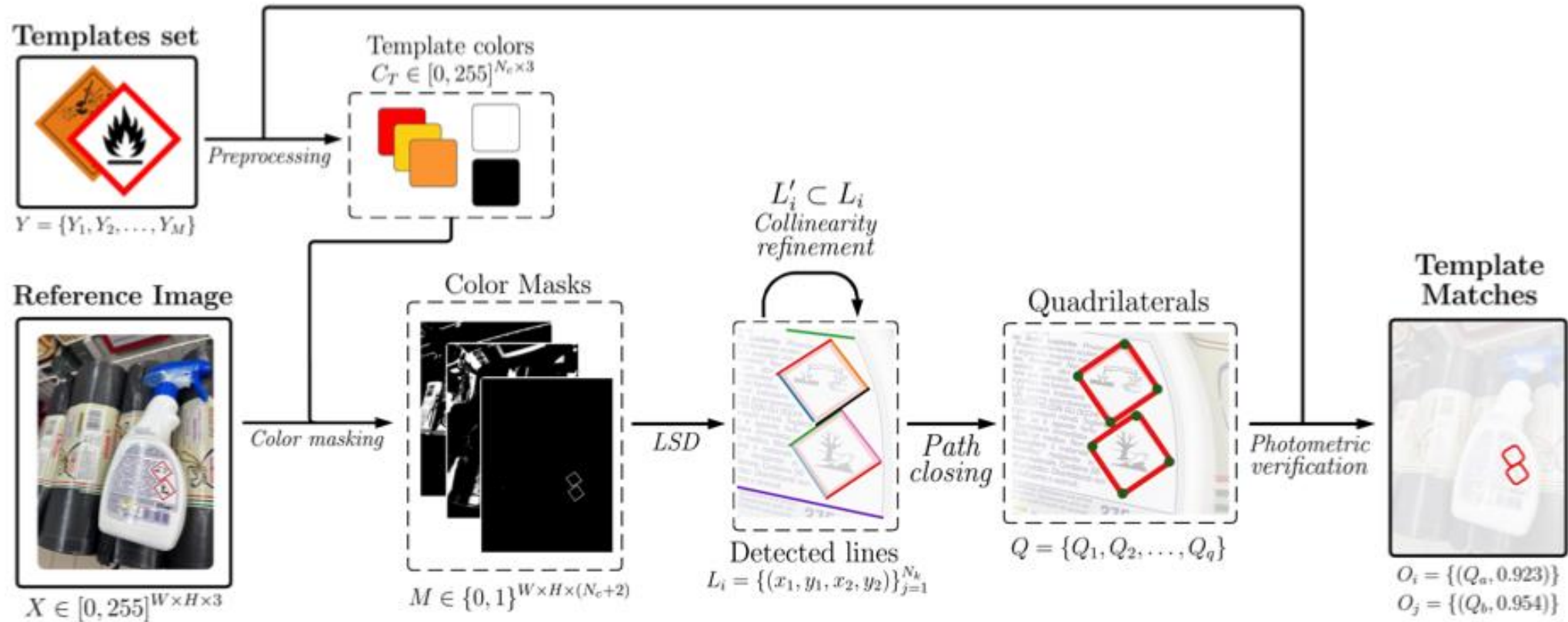
Proposed Method



Overview of the method

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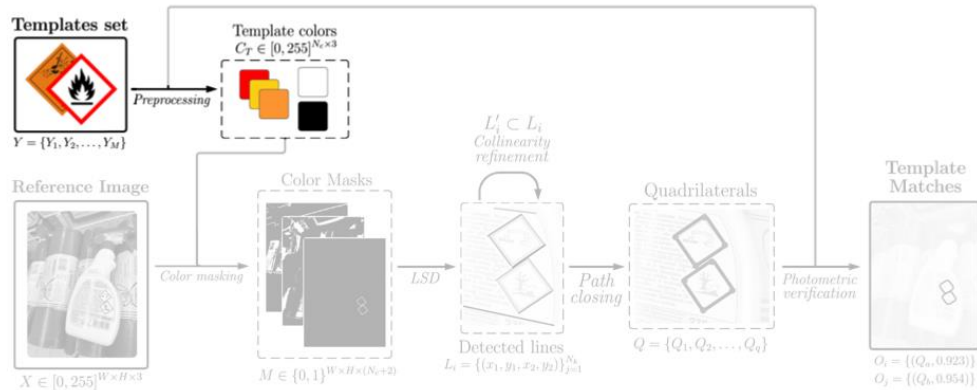
The pipeline



Then you can illustrate the method step by step

7

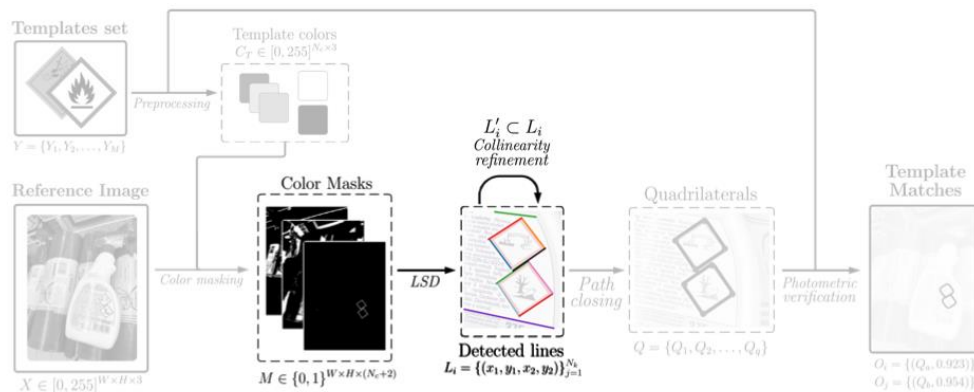
Template preprocessing



You can describe the pipeline step by step

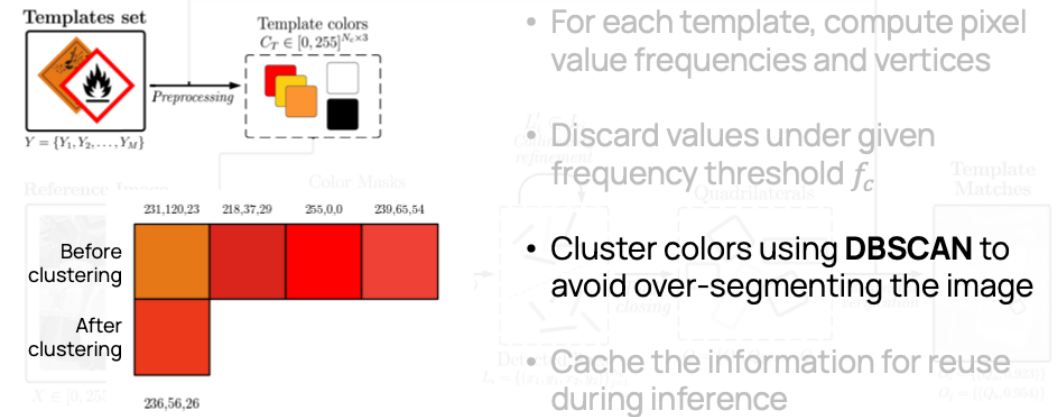
9

Line detection and refinement



7

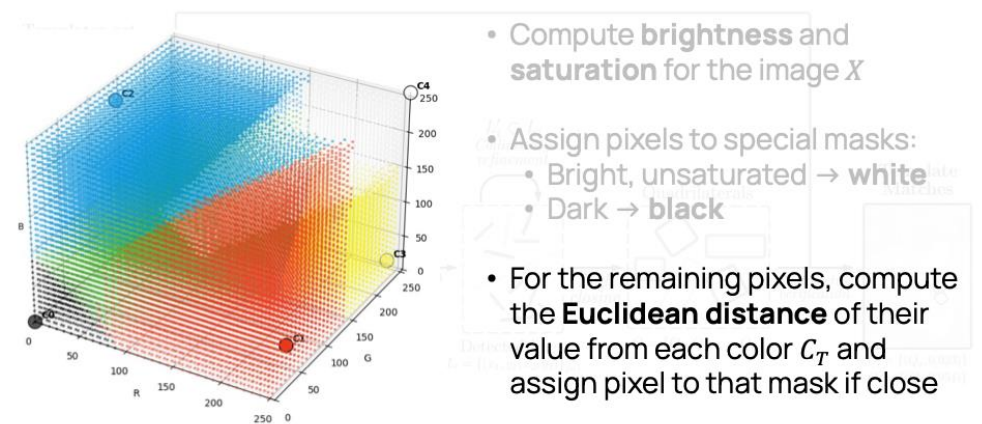
Template preprocessing



You can delve into details if they are relevant/interesting

8

Color-based mask extraction



Experiments

Organize experiments according to your claims and findings: it is not a report where you must list them all.

Present plots and results if you have an interesting comment over them. Leave to the supplementary extra stuff that can be helpful for answering question

Steer Audience Attention!

Use bold / underline with statistical meaning

Table 2. FPR/AUC achieved by the considered methods with $K = 16$ bins and batches of $\nu = 128$ points. We report the average ranking with respect to the AUC and the p -value of the Nemenyi test. For each dataset, we mark in bold the AUCs of the best-performing method, and underline them when found to be significantly different from the second best-performing.

	d	QT (w/o PCA)	QT (w/ PCA)	KQT (Euclidean)	KQT (Mahalanobis)	KQT (Weighted Maha.)	ElkM	SPLL (C=3)	PCA-SPLL (C=3)	DT (w/o PCA)	DT (w/ PCA)
unimodal	4	4.83%/ 0.96	4.81%/ 0.98	4.86%/ 0.95	4.82%/ 0.99	4.83%/ 0.99	4.82%/ 0.87	5.46%/ 1.00	5.92%/ 0.99	7.84%/ 0.79	7.75%/ 0.81
bimodal	4	4.80%/ 0.90	4.81%/ 0.93	4.80%/ 0.90	4.81%/ 0.95	4.80%/ <u>0.97</u>	4.82%/ 0.82	5.53%/ 0.92	6.02%/ 0.90	7.65%/ 0.75	7.62%/ 0.77
nino	5	5.04%/ 0.84	4.99%/ 0.91	5.00%/ 0.61	5.02%/ 0.90	5.01%/ <u>0.92</u>	4.83%/ 0.53	6.14%/ 0.82	7.69%/ 0.84	7.55%/ 0.73	7.57%/ 0.58
protein	9	4.97%/ 0.90	4.98%/ 0.98	4.97%/ 0.62	4.98%/ 0.99	5.03%/ <u>0.99</u>	4.88%/ 0.51	13.15%/ 0.92	8.42%/ 0.95	7.65%/ 0.70	7.64%/ 0.59
spruce	10	4.81%/ 1.00	4.83%/ 1.00	4.82%/ 0.60	4.84%/ 1.00	4.90%/ 1.00	4.86%/ 0.51	11.43%/ 1.00	11.56%/ 1.00	7.56%/ 1.00	7.57%/ 1.00
lodgpole	10	4.83%/ 1.00	4.82%/ 1.00	4.85%/ 0.65	4.80%/ 1.00	4.90%/ 1.00	4.92%/ 0.51	10.78%/ 1.00	10.89%/ 1.00	7.60%/ 1.00	7.58%/ 1.00
credit	28	4.83%/ 0.70	4.96%/ 0.87	4.89%/ 0.60	4.85%/ 0.78	5.06%/ 1.00	4.96%/ 0.51	8.67%/ 0.60	16.06%/ 0.66	7.63%/ 0.69	7.59%/ 0.82
insects (1 $\rightarrow$ 2)	33	4.92%/ 1.00	4.93%/ 0.96	4.91%/ 0.96	4.93%/ 0.97	5.19%/ 0.99	4.93%/ 0.84	5.90%/ 0.81	6.48%/ 0.87	7.57%/ 1.00	7.60%/ 1.00
insects (2 $\rightarrow$ 3)	33	4.93%/ 0.99	4.91%/ 1.00	4.92%/ 1.00	4.96%/ 1.00	5.25%/ 1.00	4.96%/ 0.96	5.54%/ 1.00	6.16%/ 1.00	7.60%/ 1.00	7.59%/ 1.00
insects (3 $\rightarrow$ 4)	33	4.92%/ 0.98	4.89%/ 0.90	4.90%/ 0.90	4.88%/ 0.94	5.22%/ 0.99	4.89%/ 0.83	6.09%/ 0.75	6.69%/ 0.74	7.59%/ 1.00	7.54%/ 1.00
insects (4 $\rightarrow$ 5)	33	4.92%/ 1.00	4.95%/ 1.00	4.91%/ 1.00	4.92%/ 1.00	5.25%/ 1.00	4.91%/ 0.95	5.48%/ 1.00	6.01%/ 1.00	7.63%/ 1.00	7.56%/ 1.00
insects (5 $\rightarrow$ 6)	33	4.91%/ 1.00	4.90%/ 0.97	4.90%/ 0.98	4.92%/ 0.99	5.26%/ 1.00	4.90%/ 0.96	5.86%/ 0.98	6.19%/ 0.98	7.61%/ 1.00	7.63%/ 1.00
sensorless	48	4.84%/ 0.86	5.01%/ 1.00	4.82%/ 0.54	5.01%/ 1.00	7.42%/ 1.00	4.93%/ 0.50	4.33%/ 1.00	4.83%/ 1.00	7.55%/ 0.74	7.58%/ 0.60
particle	50	4.85%/ 0.89	4.87%/ 0.93	4.81%/ 0.55	4.94%/ 0.98	5.80%/ <u>0.99</u>	4.84%/ 0.51	5.93%/ 0.84	6.07%/ 0.90	7.52%/ 0.80	7.60%/ 0.54
Average Ranking		5.24	4.93	7.08	3.82	2.98	9.37	5.57	5.34	5.11	5.56
Nemenyi p -value		$< 10^{-16}$	$< 10^{-16}$	$< 10^{-16}$	$< 10^{-16}$	-	$< 10^{-16}$	$< 10^{-16}$	$< 10^{-16}$	$< 10^{-16}$	$< 10^{-16}$

Steer Audience Attention!

Highlight important columns you want to comment on. Include comments

Table 2. FPR/AUC achieved by the considered methods with $K = 16$ bins and batches of $\nu = 128$ points. We report the average ranking with respect to the AUC and the p -value of the Nemenyi test. For each dataset, we mark in bold the AUCs of the best-performing method, and underline them when found to be significantly different from the second best-performing.

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e.g. claim: “KQT can be used instead of PCA preprocessing, achieving the best results”

Steer Audience Attention!

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e.g. claim: “In high dimension Covariance Matrices are poorly conditioned, FPR not controlled”

Conclusions

- Wrap up findings
- Mention future work
- Emphasize impact of your findings

Layout and formatting

Layout of the presentation

Use power point!

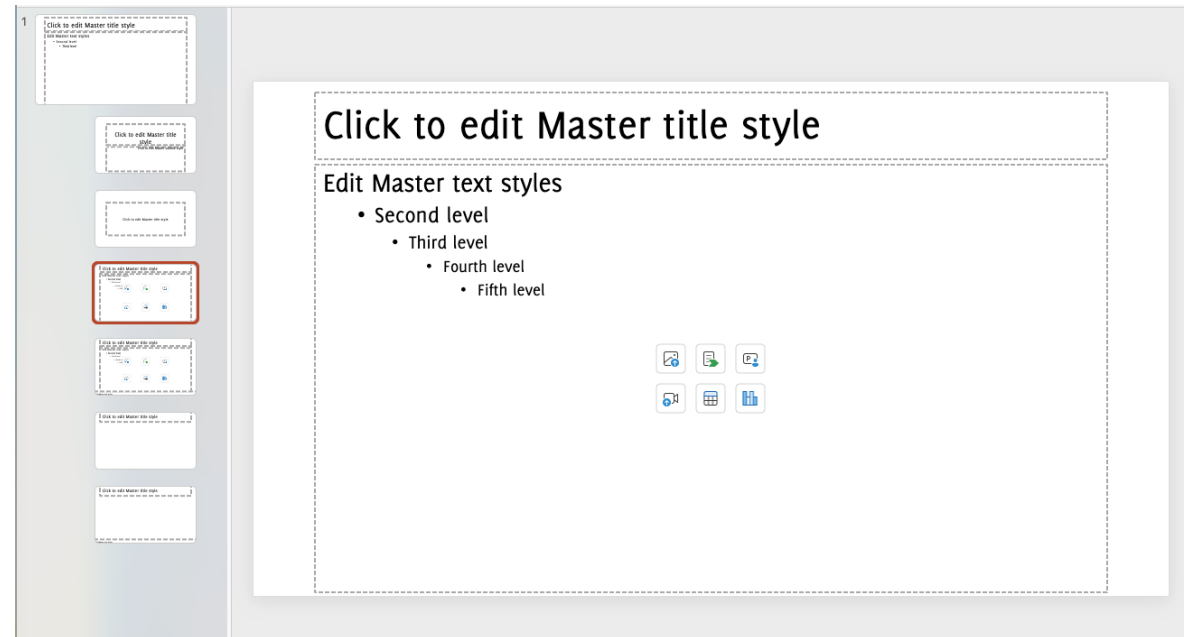
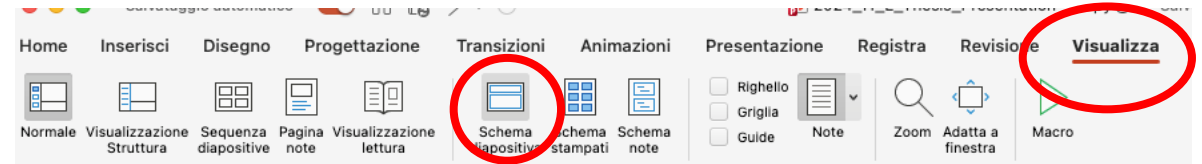
There is no specific template or master for the ppt.

Use a master for all your slide. You can also use a clean one, like this.

Make sure the slides are in **16:9 format**.

- Add slide number (very useful for handling questions / comments)

No need of «Outline» slide, save time and grab audience attention with a better start!



First Slide

There is a lot of flexibility, choose the one you prefer but, it **has to include**

- your name
- thesis title
- date
- your advisor and co-advisor name
- any collaborating entity/partner

Specifically mention if the thesis work was done during an internship (include the logo)





Enrico Barbieri 970131

Pose and camera calibration from cylinders via a silhouette based homotopy continuation



Politecnico di Milano

Relatore: Prof. Luca Magri

Correlatore: Prof. Giacomo Boracchi

Use Section Slides

Section Slides

- Give an illustrative title like «Experiments»
- No need to use «Core Contribution» or «proposed solution» as title, you can use a more specific name (e.g. the name of your solution, or the key idea in few words)
- No need to have your second slide as a «section slide» titled «Introduction»

Texts

It is good to have text in the slide.

Do not present slides with only figures and illustration.

- Audience might be lost and use text to catch up

Avoid over-itemization https://www.edwardtufte.com/bboard/q-and-a-fetch-msg?msg_id=0001yB

Make sure that:

- there are no typos.
- font size is never below 20, 24 as a standard.
- Your text is consistent in terms of typesetting (spaces, bullets need to be consistent)
- Your slides conform to the master, to prevent unusual formats

Make sure slide titles are informative... **do not ignore slide title**, people are looking at that always.

Too verbose for a slide

Introduction

Diffusion models

In figure: example of deblurring using our method.

Blurred image on the left and our reconstruction on the right.



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MILANO 1863



Generative models are models trained to be used to generate data with features learnt from the training set.

The state of the art, a few years back, were GANs. Recently, diffusion models were proven to have comparable performance in image generation and less training problems.

Also in image deblurring GANs are the most used approach. We show, for the first time, that diffusion models are a valid alternative to GANs, for image deblurring.

Compact slide text

Introduction

Diffusion models

In figure: example of deblurring using our method.

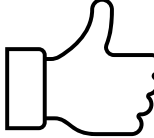
Blurred image on the left and our reconstruction on the right.



- Generative Models (GANs, Diffusion Models) have a very high potential in image restoration
- Diffusion Models are more stable than GANs in image generation.
- We are the first to train DM to perform image deblurring

Make clear your contribution!

- Formulate your text in a concise and direct form.
- Make sure you make clear what your contribution is!

 *We quantize models to 8 bit integers with scaling.. as in [x]
Our quantized model runs on a STM32*

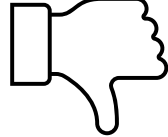
Quantization

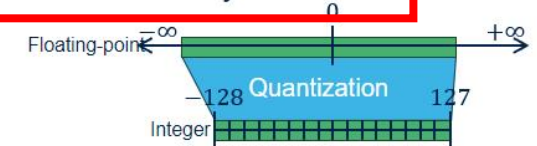
The proposed network models operate using real numbers, stored in floating-point variables.

- The networks need too many resources to be implemented on a low-power embedded device.

The post-training quantization procedure converts a trained model's data to use integer representations instead of floating-point ones.

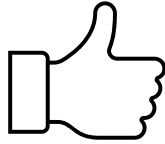
- It estimates the dynamics of the variables.
- It shifts and scales the variables to make them fit in fixed size integer variables.

 The quantized models can fit and run on low-power embedded systems.



Make clear your contribution!

- Formulate your text in a concise and direct form.
- Make sure you make clear what your contribution is!

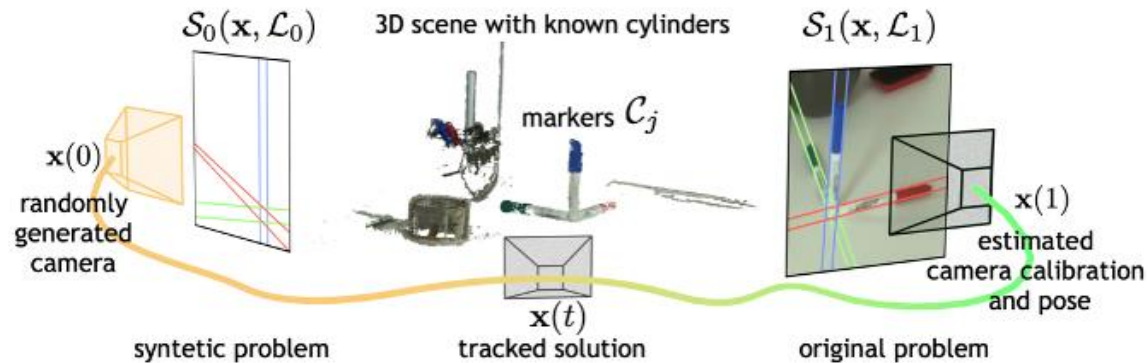


CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION

Equation arising from quadric geometry are quartic equations, each with multiple solutions. The **problem** can be **projected** into a higher space in which **is** linear but is both **computationally intensive**, and **geometrically inconsistent**.

My solution aims to solve both issues by deforming an easy, known version of the **problem** into the hard, unknown, **target problem**.

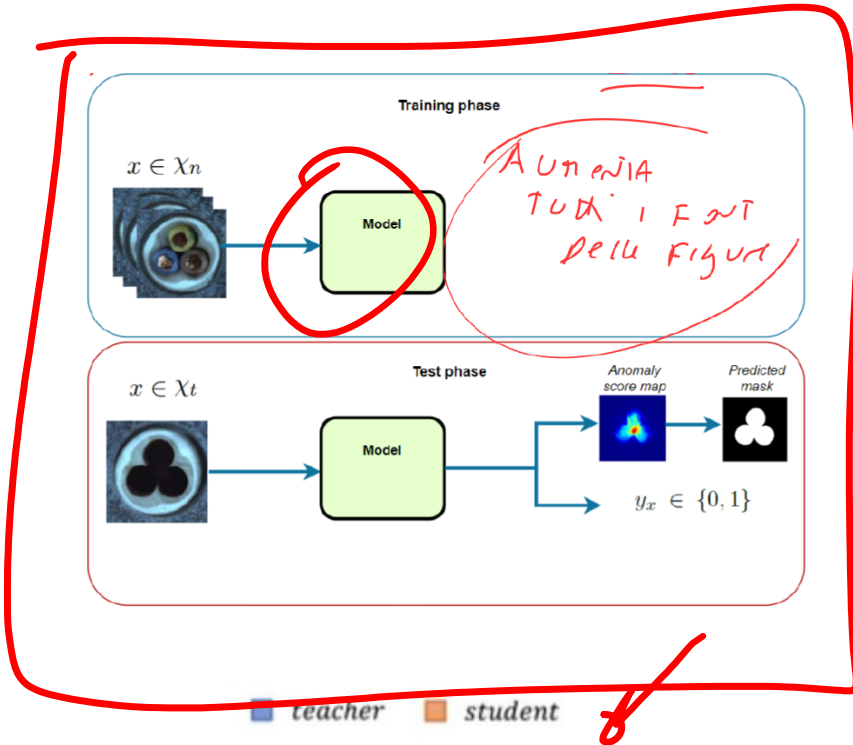
$$\begin{cases} l^h{}^T P Q^* P^T l^h = 0 \\ l^h{}^T P V = 0 \end{cases}$$



Illustrations

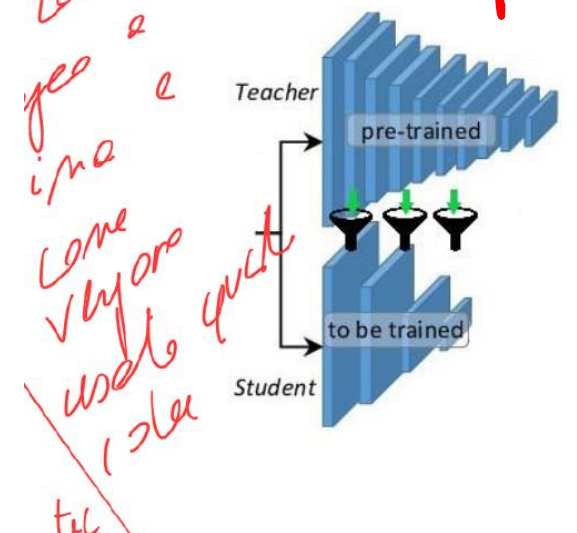
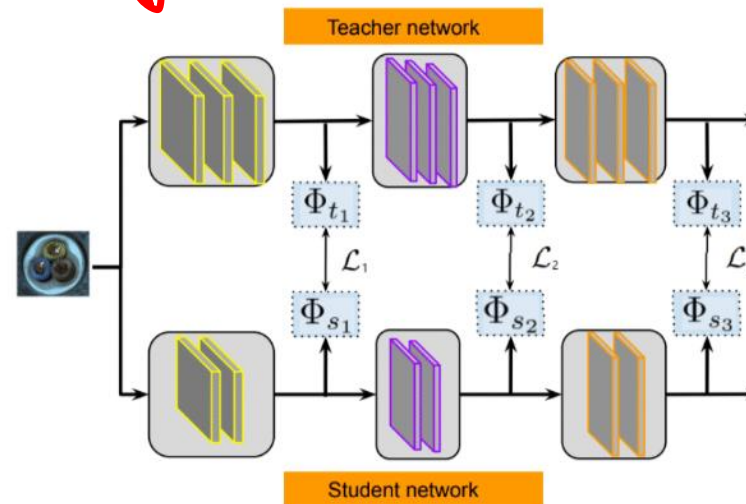
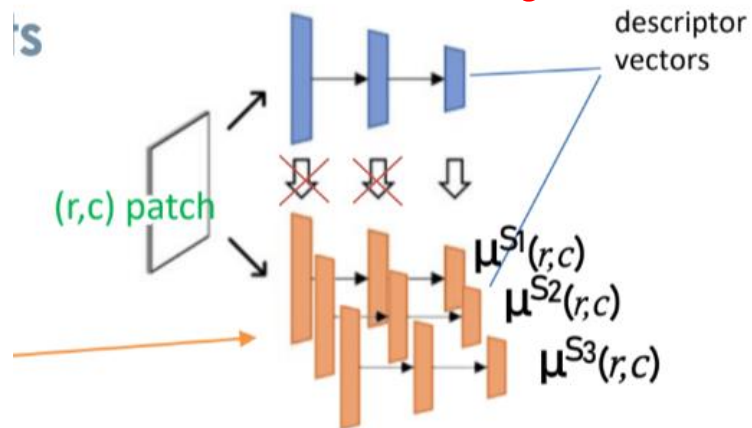
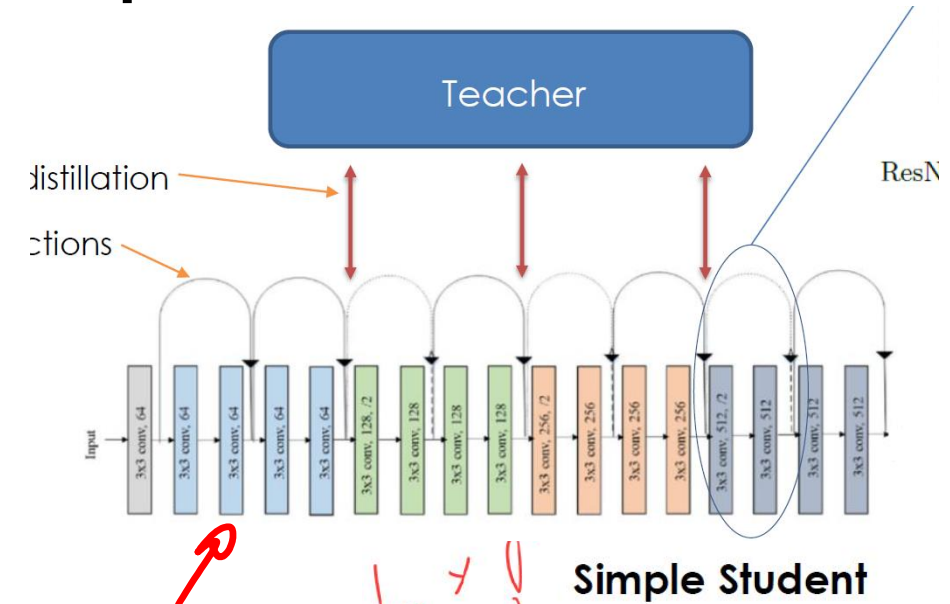
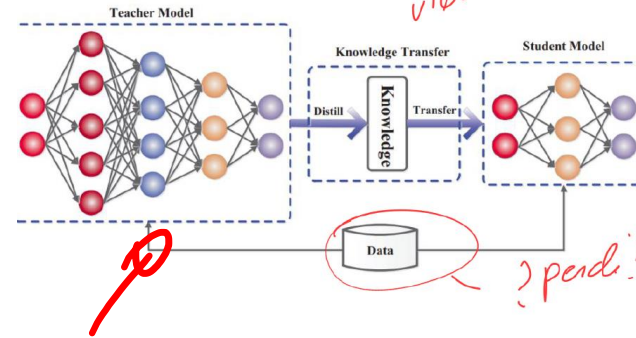
- Use **nice illustrations** and graphics! Draw at the blackboard (or tablet) the picture you need to describe in a glimpse to a colleague of yours the solution: this is the graphics you need in your slides.
- **Always put labels** to axis and titles to your plots
- Always cite image sources
- Preserve aspect ratio!!!!
- Don't use generated images (Hallucinations in schemes and references are not acceptable)
- **Use consistent style in the illustrations**

Examples (from the same draft presentation)



from strong teacher's outputs

for visual anomaly detection?



Last slide

This is going to stay longer, use it to show a plot / image / video / scheme of your algorithm to let people better understand



Avoid Repetitions!

No repetitions allowed in any level:

- Text: make sure to mention all the concepts needed once

Identity recognition

Identity recognition is the task of determine the identity of a subject.

- It answer the question “Who is this?”

... a few slides after ...

Deep identity recognition

The process of identity recognition (i.e., to tell if an input sample belong to an enrolled identity) relies on two components:

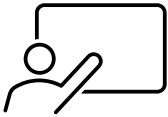
Avoid Repetitions!

No repetitions allowed in any level:

- Text
- Slides
- Content: do not introduce basic concepts “out of context” and then go back to these to illustrate your solution.

Do the other way round: introduce the concept for your solution, mentioning where it is from!

The Live Show

- For any reason, **never read**, never learn your speech by heart.
-  - Never look at the screen behind you... just look at the audience or at your laptop. Try to **grab audience attention**. Good to look at someone how is looking at you
- Use **pointer** (the one in ppt is ok), do not indicate on slides, as you need to  turn
-  - Make sure you **do not exceed the time** scheduled.
- Give your talk in **Italian** (if you are a native Italian speaker). You need a strong argument for shifting to English.
 - Audience will better understand Italian
 - You can better express your **ideas** and **passion**  

Remember “pros” will grade you. **Don't be too shallow in your presentation**



Here are some very general guidelines

How to give a brilliant presentation:

<https://www.microsoft.com/en-us/research/academic-program/give-great-research-talk/>

Please watch this video carefully and take inspiration to prepare yours.

Let's organize the work

Schedule

1. **Find your granny:** make tell them your Introductory slides
2. Prepare your slides **for pros**

Teams:

- Camilo – Marco
- Christian – Mattia
- Simone – Angelo - Francesco

Prepare your slides autonomously, give the talk to him. Gather feedback, modify slides. Iterate at least twice.

Act as a very attentive audience! Give feedback to the pro presenting his work to you.

Schedule

Send slides to your advisor by some sharing link (dropbox, googleDrive, OneDrive, WeTransfer) not to fill our inbox.

- If you had a brilliant PhD student supervising you, please check with him first.
- Make sure you can fit the 15' presentation
- Send us slides as **final...** the best you can do. Most probably there won't be any iteration and you will receive my feedback late!

Deadline: Wed 18th

Once completed, send me also the final version you are going to present.